LECTURE 17

31 October 2023 11:41

$\int_2^3 x^2 \, dx$ $\qquad \int_0^1 \frac{x^8}{1+x^8} \, dx$ $\qquad f(x) = \begin{cases} x+2 & 0 < x \le 1 \\ -2x+3 & 1 \le x \le 2 \end{cases}$

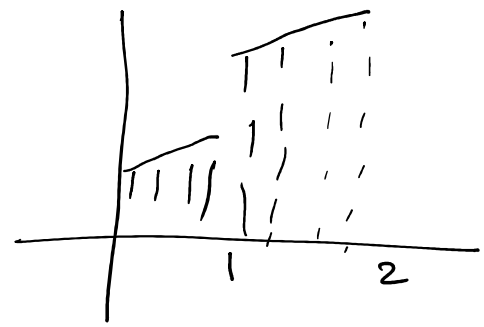
1) f is conti. on $[a,b] \Rightarrow f \to$ R-integrable.

$f(x) = 1 - \frac{1}{1+x^8}$ $\qquad f'(x) = -\frac{1}{(1+x^8)^2} \cdot 8x^7 \le 0$ $\quad \forall x \in (0,1)$

2) f is monotone on $[a,b] \Rightarrow f$ is R-integrable.

Eg $f(x) = \begin{cases} 1 & x \in \mathbb{Q} \\ -1 & x \in \mathbb{I} \end{cases}$

not R-integrable.



3) f has finite no. of dis. conti. on $[a,b] \Rightarrow f \to$ R-integr.

Fundamental thrm of calculus: 1) f is conti. on $[a,b]$

2) $\exists$ F conti. on (a,b) s.t. $F'(x) = f(x)$

3) $f \to$ R-integrable on $[a,b]$

then $\int_a^b f(x) \, dx = F(b) - F(a)$

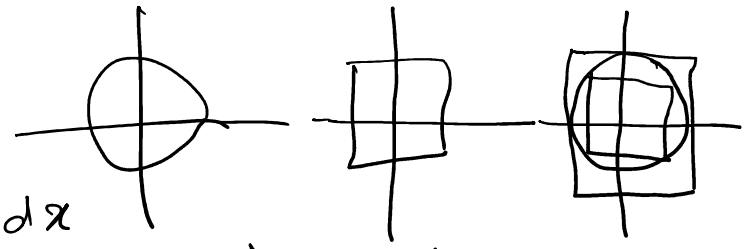
Eg $\int_2^5 x^2 \, dx =$ $\qquad f(x) = x^2$ $\qquad F(x) = \frac{x^3}{3}$

$F(5) - F(3) = \frac{5^3}{3} - \frac{2^3}{3} = ?$

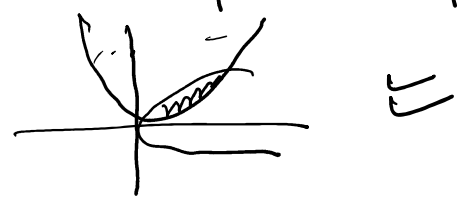
$\int x\sqrt{1+x^2}$

Area b/w two curves:

$y = f(x)$, $y = g(x)$ $\quad x = a$, $x = b$



$$A.B.C = \int_a^b \{f(x) - g(x)\}\, dx$$



Eg: 1) $y = x^2$ $\quad y = 2x + 3$

Step 1: Find intersection: $x^2 = 2x + 3$

$x^2 - 2x - 3 = 0$

$x = -1, 3$

$$\int_{-1}^{3} x^2 - (2x+3)\, dx$$

$=$

2) $y^2 = x$ & $x^2 = y$

P.O.I. 0 & 1

$$\int_0^1 (\sqrt{x} - x^2)\, dx = 1/3$$